

Tung Yan Zuo, Darren

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Personal Website: darren-tung.github.io/Profile/

PROFESSIONAL EXPERIENCE

Software Engineering Intern | Rhaeos, Inc. | United States, Illinois, Chicago Aug 2025 - Jul 2026

- **Re-architected** entire **Bluetooth Low Energy (BLE)** Manager Submodule infrastructure using **TypeScript, AWS & React Native** to facilitate seamless integration, scalability, and usage within all Rhaeos **mobile applications**
- Transitioned core product from its predecessor to a clinically ready state utilising **TypeScript, AWS, NodeJS, React Native, Xcode & Python** directly supporting 2+ clinical studies & its path to FDA 510k approval
- **Created** the Validator App using **TypeScript, React Native, & BLE** to verify hardware functionality which established a robust **QA** protocol projected to **save hundreds of hours** of manual testing
- **Architected** and built foundational framework for the Rhaeos OTP forgot-password system utilising **AWS (Lambda, Cognito, Simple Email System, EventBridge, CloudTrail)**

Software Engineering Intern | Y3 Technologies | Singapore, SG May 2025 - Aug 2025

- **Led automation** project to convert **Jira Tickets** to carefully curated test cases, which will be converted to Excel files to be sent to testing team, using **Python, Anthropic AI & Prompt Engineering**
- Set up & facilitated **CI/CD deployment pipeline** to company's server utilising **GitHub Actions, Bash, Git & Docker**
- Created a dynamic **UI** for non-developers to interact with core functionality, minimising interaction with **CLI** using **Vite, Tailwindcss, ReactJS, Typescript & Shadcn**, while having a separate administrator window to view logs
- Projected to impact **over 100** staff members & **save Y3 Technologies over \$10000** annually

Full Stack Software Engineering Intern | Backbone Pte. Ltd. | Singapore, SG Sep 2024 - May 2025

- Led full-stack redevelopment of Backbone's corporate website utilising **Next.js, Strap, Shadcn, TypeScript, Docker & Tailwind CSS**, delivering enhanced performance, maintainability, & consistent design across all pages with **CI/CD pipeline** via **GitHub Actions** for automated deployment to **Microsoft Azure**
- Engineered **Python** automation solution that converts client financial statements into dynamic business rules stored in **MySQL** database, reducing manual processing time by **80%** & **eliminating data entry errors**
- Architected seamless data export pipeline to **Microsoft Azure**, enabling automatic integration of custom rule sets with Backbone's core platform & delivering personalised client solutions with zero manual intervention

Administrative & Training Specialist | Ministry of Defence | Singapore, SG Oct 2021 - Aug 2023

- Built Telegram Bots utilising **Google App Script & Microsoft Excel VBA** for automation purposes
- Impacted over **1000** recruits & staff, improved work efficiency tremendously & **saved 80% of time**
- **Awards:** ALTI Exemplary Instructional Leader (Innovation) Award

Software Engineering Intern | Defence Science & Technology Agency | Singapore, SG Mar 2021 - Jun 2021

- Collaborated on an AI Chatbot using **Rasa Open Source & Python** to allow commanders to make quick decisions
- Played a role in advancing an MLOps project which employs **Docker & Harbor** to advance AI capabilities

TECHNICAL SKILLS

- **Programming Languages:** C, C++, C#, Python, JavaScript, HTML, Java, R, TypeScript
- **Frameworks & Technologies:** ReactJS, NodeJS, WebRTC, React Native, OpenCV, Shadcn, Vite, NextJS, XCode
- **Databases & Cloud Services:** MySQL, MongoDB, Firebase, Microsoft Azure, Amazon Web Services, SQLancer
- **Tools & DevOps:** Git, GitHub Actions, Docker, Google App Scripts, Microsoft VBA, R Studios

CERTIFICATION

Amazon Web Services (AWS) Certified AI Practitioner Certificate	Mar 2026
Amazon Web Services (AWS) Certified Solution Architect - Associate	Aug 2025
Scrum Alliance Certified ScrumMaster (CSM)	Jun 2025
Amazon Web Services (AWS) Certified Cloud Practitioner Certificate	Jun 2025
WILEY Certified Data Analyst	Apr 2022

EDUCATION

- National University of Singapore | Singapore, SG** **Aug 2023 - Dec 2026**
- **Bachelor of Computing in Computer Science (Honours) | Specialisation:** SWE & Networking
 - **Relevant Courses:** Data Structures & Algorithms (**Java**), Introduction to AI & ML (**AI**)
- University of Illinois Chicago | United States, Illinois, Chicago** **Aug 2025 - Jul 2026**
- Attended an entrepreneurship curriculum under NUS Overseas College (NOC)
- Ngee Ann Polytechnic | Singapore, SG** **Apr 2018 - May 2021**
- **Diploma in Engineering Science | GPA:** 3.9266/4.0
 - **Relevant Courses:** Data Structures & Algorithms (**C++**), Fundamentals of Obj-oriented Programming (**C#, OO**)
 - **Awards:** DSTA Polytechnic Engineering Scholarship, Director's List (Year 2.1), Director's List (Year 1.2)

PROJECTS

- SQLancer Benchmark | User Interface Team | NUS Software Engineering Project** **Jan 2025 - May 2025**
- **Led a team of 3 developers** to design & implement a web app using **Next.js, Shadcn, TypeScript, ReactJS & NodeJS**, which will allow non-developers to use our product with no issues
 - Collaborated with Metrics team & facilitated integration of frontend & backend processes with **unit & integration testing, SQLancer & bash scripts**
- TalentSG Web App | NUS Software Engineering Project** **Oct 2024 - Dec 2024**
- Designed web app on **Canva** & integrated it utilising **JavaX, CSS, CI-CD Pipelining & Java**
 - **Led a team of 5 developers** & conducted weekly **sprints & scrum** meetings while leveraging **Agile & Git**
- Health Screening 2024 | Self-Sourced Project | Public Health Service (PHS)** **May 2024 - Aug 2024**
- Impacted over **1000** volunteers & participants who underwent a health screening by PHS
 - Developed & updated web-based queue system & data storage with **JavaScript, ReactJS & MongoDB**
 - **Led a team of 4 developers** & streamlined its development leveraging **Agile & Git**
 - Responsible for communicating with PHS committee & delegating tasks to team members
- UniFyTime | Independent Software Development Project (Orbital - Advance)** **May 2024 - Aug 2024**
- Worked in pairs to propose & develop a web-based calendar system using **JavaScript, ReactJS & Firebase API**
 - Designed user interface (**UI**) for optimum user experience (**UX**), structuring information architecture for intuitive navigation aligning with modern designs & best practices, real-time synchronisation with **Firebase & Firestore**
 - Coordinated effective communication within team through daily developmental sprints utilising **Agile & Git**
 - Encompassed Telegram Bot Integration using **JavaScript & Telegraf.js**

LEADERSHIP EXPERIENCE

- Treasurer (Grassroots Leader) | Zhenghua Community Club Youth Network** **Oct 2016 - Present**
- Oversee claims & expenditures, approve reimbursements, & ensure accurate financial documentation
 - Review SOA & transactions before meetings, maintaining transparency & accountability in committee funds
 - Organised numerous community events that reached out to over 300 residents in the Zhenghua community
 - Volunteered in various community events, such as a weekly food rescue event
 - **Awards:** People's Association Certificate of Appreciation (5 Years Long Service Award)

REFEREES

Scott Odland

Head of Software at Rhaeos Inc.

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Arunraj s/o Chidambaram

Chairman of Zhenghua Community Club Youth Network, Holland-Bukit Timah GRC

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